

Riemann Integration

From Partitions to Integrability

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Partition of an Interval

Definition: A partition of $[a, b]$ is a finite ordered set:

$$P = (x_0, x_1, \dots, x_n) \quad \text{with} \quad a = x_0 < x_1 < \dots < x_n = b$$

Example

For $[0, 1]$:

- $P = (0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1)$
- $Q = (0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{7}{8}, 1)$

Notation: $\mathcal{P}[a, b]$ = family of all partitions of $[a, b]$

Upper and Lower Sums

For bounded $f : [a, b] \rightarrow \mathbb{R}$ and partition $P = (x_0, \dots, x_n)$:

$$M_r = \sup_{x \in [x_{r-1}, x_r]} f(x)$$

$$m_r = \inf_{x \in [x_{r-1}, x_r]} f(x)$$

Darboux Sums

$$U(P, f) = \sum_{r=1}^n M_r (x_r - x_{r-1}) \quad (\text{Upper Sum})$$

$$L(P, f) = \sum_{r=1}^n m_r (x_r - x_{r-1}) \quad (\text{Lower Sum})$$

Properties of Darboux Sums

Theorem

For any partition P :

$$m(b-a) \leq L(P, f) \leq U(P, f) \leq M(b-a)$$

where $M = \sup_{[a,b]} f(x)$, $m = \inf_{[a,b]} f(x)$

Proof.

Since $m \leq m_r \leq M_r \leq M$ for all r , we have:

$$m \sum (x_r - x_{r-1}) \leq \sum m_r (x_r - x_{r-1}) \leq \sum M_r (x_r - x_{r-1}) \leq M \sum (x_r - x_{r-1})$$



Refinement of Partitions

Definition: Q is a refinement of P if $P \subseteq Q$

Example

If $P = (0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1)$ and
 $Q = (0, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{7}{8}, 1)$
then Q is a refinement of P

Lemma

If Q refines P , then:

$$L(P, f) \leq L(Q, f) \leq U(Q, f) \leq U(P, f)$$

Upper and Lower Integrals

Definition

$$\underline{\int_a^b} f = \sup\{L(P, f) : P \in \mathcal{P}[a, b]\} \quad (\text{Lower Integral})$$

$$\overline{\int_a^b} f = \inf\{U(P, f) : P \in \mathcal{P}[a, b]\} \quad (\text{Upper Integral})$$

Theorem

For any bounded function f :

$$\underline{\int_a^b} f \leq \overline{\int_a^b} f$$

Definition

f is Riemann integrable on $[a, b]$ if

$$\underline{\int_a^b} f = \overline{\int_a^b} f$$

The common value is denoted $\int_a^b f$

Notation

- $\mathcal{R}[a, b]$: Riemann integrable functions on $[a, b]$
- $\mathcal{B}[a, b]$: Bounded functions on $[a, b]$
- $\mathcal{R}[a, b] \subset \mathcal{B}[a, b]$ (proper subset)

Example 1: Constant Function

Let $f(x) = c$ on $[a, b]$

For any partition P :

$$U(P, f) = c(b - a)$$

$$L(P, f) = c(b - a)$$

Therefore:

$$\underline{\int_a^b f} = \overline{\int_a^b f} = c(b - a)$$

So $f \in \mathcal{R}[a, b]$ and $\int_a^b f = c(b - a)$

Example 2: Dirichlet Function

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

For any partition P :

$$U(P, f) = 1$$

$$L(P, f) = 0$$

Therefore:

$$\underline{\int_0^1} f = 0 \neq 1 = \overline{\int_0^1} f$$

So $f \notin \mathcal{R}[0, 1]$

Example 3: Identity Function

Let $f(x) = x$ on $[a, b]$

Take partition P_n with equal subintervals:

$$h = \frac{b-a}{n}, \quad x_r = a + rh$$

Since f is increasing:

$$U(P_n, f) = (b-a)a + \frac{1}{2}(b-a)^2\left(1 + \frac{1}{n}\right)$$

$$L(P_n, f) = (b-a)a + \frac{1}{2}(b-a)^2\left(1 - \frac{1}{n}\right)$$

Therefore:

$$\int_a^b f = \overline{\int_a^b f} = \frac{b^2 - a^2}{2}$$

Key Theorems

Theorem

For any two partitions P, Q :

$$L(P, f) \leq U(Q, f)$$

Theorem (Integrability Criterion)

$f \in \mathcal{R}[a, b]$ iff for every $\epsilon > 0$, there exists partition P such that:

$$U(P, f) - L(P, f) < \epsilon$$

Summary

Next Steps

Study properties of integrable functions and the Fundamental Theorem of Calculus

Let $[a, b]$ be a closed and bounded interval and $c \in \mathbb{R}$. A function $f : [a, b] \rightarrow \mathbb{R}$ is defined by

$$f(x) = c, \quad x \in [a, b].$$

Claim: $f \in \mathcal{R}[a, b]$ (i.e., f is Riemann integrable).

Step 1: Boundedness

Since $f(x) = c$ for all $x \in [a, b]$,

$$\sup f(x) = \inf f(x) = c.$$

Thus, f is bounded on $[a, b]$.

Step 2: Partition and Upper Sum

Take a partition $P = (x_0, x_1, \dots, x_n)$ of $[a, b]$, where

$$a = x_0 < x_1 < \dots < x_n = b.$$

For each subinterval $[x_{r-1}, x_r]$:

$$M_r = \sup f(x) = c, \quad m_r = \inf f(x) = c.$$

Hence,

$$U(P, f) = \sum_{r=1}^n M_r(x_r - x_{r-1}) = c(b - a).$$

Step 3: Lower Sum

Similarly,

$$L(P, f) = \sum_{r=1}^n m_r(x_r - x_{r-1}) = c(b - a).$$

Thus, for every partition P ,

$$U(P, f) = L(P, f) = c(b - a).$$

Conclusion

Since the upper and lower sums are equal for every partition,

$$\int_a^b f(x) dx = c(b - a).$$

Hence, $f(x) = c \in \mathcal{R}[a, b]$.

Definition

A function f is defined on $[0, 1]$ by

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational,} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

Claim: f is not Riemann integrable on $[0, 1]$.

Step 1: Boundedness

Clearly,

$$0 \leq f(x) \leq 1, \quad \forall x \in [0, 1].$$

Thus, f is bounded on $[0, 1]$.

Step 2: Partition and Sup/Inf Values

Take a partition $P = (x_0, x_1, \dots, x_n)$ of $[0, 1]$, where

$$0 = x_0 < x_1 < \dots < x_n = 1.$$

For each interval $[x_{r-1}, x_r]$:

- $\sup f(x) = 1$ (since rationals exist in every interval), - $\inf f(x) = 0$ (since irrationals exist in every interval).

Thus,

$$M_r = 1, \quad m_r = 0 \quad \text{for all } r = 1, 2, \dots, n.$$

Step 3: Upper and Lower Sums

Upper sum:

$$U(P, f) = \sum_{r=1}^n M_r(x_r - x_{r-1}) = \sum_{r=1}^n (x_r - x_{r-1}) = 1.$$

Lower sum:

$$L(P, f) = \sum_{r=1}^n m_r(x_r - x_{r-1}) = \sum_{r=1}^n 0 = 0.$$

Conclusion

For any partition P of $[0, 1]$,

$$U(P, f) = 1, \quad L(P, f) = 0.$$

Since

$$\sup L(P, f) = 0, \quad \inf U(P, f) = 1,$$

and they are not equal,

$$f \notin \mathcal{R}[0, 1].$$

Lemma 11.4.2

Let a function $f : [a, b] \rightarrow \mathbb{R}$ be bounded on $[a, b]$ and P, Q be any two partitions of $[a, b]$. Then

$$L(P, f) \leq U(Q, f), \quad L(Q, f) \leq U(P, f).$$

Proof (Step 1)

Let $S = P \cup Q$. Then the partition S is a refinement of both P and Q . By previous lemmas (11.2.1 and 11.3.1),

$$L(P, f) \leq L(S, f) \leq U(S, f) \leq U(P, f),$$

$$L(Q, f) \leq L(S, f) \leq U(S, f) \leq U(Q, f).$$

Proof (Step 2)

From the above inequalities, we obtain:

$$L(P, f) \leq L(S, f) \leq U(S, f) \leq U(Q, f),$$

and

$$L(Q, f) \leq L(S, f) \leq U(S, f) \leq U(P, f).$$

Conclusion

Thus, we have shown that

$$L(P, f) \leq U(Q, f), \quad L(Q, f) \leq U(P, f).$$

Lemma 11.4.2 is proved.

Theorem 11.4.4 (Condition for Integrability)

Let a function $f : [a, b] \rightarrow \mathbb{R}$ be bounded on $[a, b]$. Then f is integrable on $[a, b]$ if and only if for each $\varepsilon > 0$ there exists a partition P of $[a, b]$ such that

$$U(P, f) - L(P, f) < \varepsilon.$$

Proof of Theorem 11.4.4 (Step 1)

Suppose $f \in \mathcal{R}[a, b]$. Then

$$\int_a^b f = \overline{\int_a^b f}.$$

Let $\varepsilon > 0$. Since $\overline{\int_a^b f}$ is the least upper bound of $\{L(P, f) : P \in \mathcal{P}[a, b]\}$, there exists a partition P' such that

$$\int_a^b f - \frac{\varepsilon}{2} < L(P', f) \leq \overline{\int_a^b f}.$$

Proof of Theorem 11.4.4 (Step 2)

Similarly, since $\overline{\int_a^b f}$ is the greatest lower bound of $\{U(P, f) : P \in \mathcal{P}[a, b]\}$, there exists a partition P'' such that

$$\int_a^b f \leq U(P'', f) < \int_a^b f + \frac{\varepsilon}{2}.$$

Let $P = P' \cup P''$. Then P is a refinement of both P' and P'' . Thus,

$$L(P, f) \leq L(P'', f) \leq U(P, f) \leq U(P'', f).$$

Proof of Theorem 11.4.4 (Step 3)

Combining inequalities:

$$\int_a^b f - \frac{\varepsilon}{2} < L(P, f) \leq U(P, f) < \int_a^b f + \frac{\varepsilon}{2}.$$

Hence,

$$U(P, f) - L(P, f) < \varepsilon.$$

Conversely, if for every $\varepsilon > 0$ there exists P such that

$$U(P, f) - L(P, f) < \varepsilon,$$

then $\underline{\int_a^b} f = \overline{\int_a^b} f$, so $f \in \mathcal{R}[a, b]$.

Theorem Statement

Theorem 11.5.1. Let a function $f : [a, b] \rightarrow \mathbb{R}$ be monotone on $[a, b]$. Then f is integrable on $[a, b]$.

We will prove the case for monotone increasing functions. The proof for decreasing functions follows similarly.

Proof: Setup

Proof. Let f be monotone increasing on $[a, b]$.

- f is bounded: $f(a) \leq f(x) \leq f(b)$ for all $x \in [a, b]$
- Let $\epsilon > 0$ be given
- Choose partition P with $\|P\| < \frac{\epsilon}{f(b) - f(a) + 1}$
- Let $P = (x_0, x_1, \dots, x_n)$, where $a = x_0 < x_1 < \dots < x_n = b$

Supremum and Infimum on Subintervals

For each subinterval $[x_{r-1}, x_r]$, define:

$$M_r = \sup_{x \in [x_{r-1}, x_r]} f(x), \quad m_r = \inf_{x \in [x_{r-1}, x_r]} f(x)$$

Since f is increasing:

$$M_r = f(x_r), \quad m_r = f(x_{r-1}) \quad \text{for } r = 1, 2, \dots, n$$

Upper and Lower Sums Difference

The difference between upper and lower sums is:

$$\begin{aligned} U(P, f) - L(P, f) &= \sum_{r=1}^n (M_r - m_r)(x_r - x_{r-1}) \\ &= \sum_{r=1}^n (f(x_r) - f(x_{r-1}))(x_r - x_{r-1}) \end{aligned}$$

Estimating the Difference

Since $x_r - x_{r-1} \leq \|P\|$ for all r , we have:

$$\begin{aligned} U(P, f) - L(P, f) &\leq \|P\| \sum_{r=1}^n (f(x_r) - f(x_{r-1})) \\ &= \|P\| (f(b) - f(a)) \end{aligned}$$

Completing the Proof

By our choice of partition:

$$\|P\| < \frac{\epsilon}{f(b) - f(a) + 1} < \frac{\epsilon}{f(b) - f(a)} \quad (\text{if } f(b) > f(a))$$

$$U(P, f) - L(P, f) < \epsilon$$

If $f(b) = f(a)$, then f is constant and trivially integrable.

Therefore, for any $\epsilon > 0$, there exists a partition P such that:

$$U(P, f) - L(P, f) < \epsilon$$

This is the Riemann criterion for integrability.

Theorem Statement

Theorem 11.5.2. Let a function $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$. Then f is integrable on $[a, b]$.

This result establishes that all continuous functions on a closed interval are Riemann integrable.

Proof: Setup

Proof.

- Since f is continuous on $[a, b]$, it is bounded (Extreme Value Theorem)
- Since f is continuous on a closed interval, it is uniformly continuous
- Let $\epsilon > 0$ be given
- By uniform continuity, there exists $\delta > 0$ such that:

$$|f(x') - f(x'')| < \frac{\epsilon}{b - a} \quad \text{for all } x', x'' \in [a, b] \text{ with } |x' - x''| < \delta$$

Constructing the Partition

- Choose a partition P with $\|P\| < \delta$
- Let $P = (x_0, x_1, \dots, x_n)$, where $a = x_0 < x_1 < \dots < x_n = b$
- Define for each subinterval $[x_{r-1}, x_r]$:

$$M_r = \sup_{x \in [x_{r-1}, x_r]} f(x), \quad m_r = \inf_{x \in [x_{r-1}, x_r]} f(x)$$

Using the Extreme Value Theorem

- Since f is continuous on each closed subinterval $[x_{r-1}, x_r]$
- By the Extreme Value Theorem, there exist points $\xi_r, \eta_r \in [x_{r-1}, x_r]$ such that:

$$f(\xi_r) = M_r, \quad f(\eta_r) = m_r$$

- Since $|\xi_r - \eta_r| \leq (x_r - x_{r-1}) \leq \|P\| < \delta$
- By uniform continuity:

$$|M_r - m_r| = |f(\xi_r) - f(\eta_r)| < \frac{\epsilon}{b - a}$$

Estimating the Difference of Sums

The difference between upper and lower sums is:

$$U(P, f) - L(P, f) = \sum_{r=1}^n (M_r - m_r)(x_r - x_{r-1})$$

$$< \frac{\epsilon}{b-a} \sum_{r=1}^n (x_r - x_{r-1}) = \frac{\epsilon}{b-a} \cdot (b-a) = \epsilon$$

Completing the Proof

- For any $\epsilon > 0$, we have found a partition P such that:

$$U(P, f) - L(P, f) < \epsilon$$

- This satisfies the Riemann criterion for integrability
- Therefore, f is integrable on $[a, b]$

Important Notes

Note 1: The class of all functions continuous on $[a, b]$ is denoted by $\mathcal{C}[a, b]$. The theorem shows that:

$$f \in \mathcal{C}[a, b] \Rightarrow f \in \mathcal{R}[a, b]$$

Note 2: $\mathcal{C}[a, b]$ is a proper subset of $\mathcal{R}[a, b]$. There exist discontinuous functions that are still Riemann integrable.

This demonstrates that continuity is a sufficient but not necessary condition for Riemann integrability.

Theorem Statement

Theorem 11.5.3. Let a function $f : [a, b] \rightarrow \mathbb{R}$ be bounded on $[a, b]$ and continuous except at a finite number of points. Then f is integrable on $[a, b]$.

This result shows that bounded functions with finitely many discontinuities are still Riemann integrable.

Proof.

- Since f is bounded on $[a, b]$, $\exists k > 0$ such that $|f(x)| < k$ for all $x \in [a, b]$
- Let f be discontinuous at m points $x_1, x_2, \dots, x_m$ with $x_1 < x_2 < \dots < x_m$
- Let $\epsilon > 0$ be given

Case 1: Discontinuities in the Interior

Assume $a < x_1 < x_2 < \cdots < x_m < b$.

- Enclose each discontinuity x_r in an interval $[x_r - \frac{\delta_r}{2}, x_r + \frac{\delta_r}{2}]$
- Choose $\delta_1 + \delta_2 + \cdots + \delta_m < \frac{\epsilon}{4k}$
- On these intervals, $M^{(r)} - m^{(r)} \leq 2k$ (since $|f(x)| < k$)

Handling the Continuous Parts

- The function is continuous on the remaining $m + 1$ subintervals:

$$\left[a, x_1 - \frac{\delta_1}{2}\right], \left[x_1 + \frac{\delta_1}{2}, x_2 - \frac{\delta_2}{2}\right], \dots, \left[x_m + \frac{\delta_m}{2}, b\right]$$

- On each continuous subinterval, we can find partitions P_k such that:

$$U(P_k, f) - L(P_k, f) < \frac{\epsilon}{2(m+1)} \quad \text{for } k = 1, 2, \dots, m+1$$

Combining the Partitions

Let $P = P_1 \cup P_2 \cup \cdots \cup P_{m+1}$. Then:

$$\begin{aligned} U(P, f) - L(P, f) &= \sum_{k=1}^{m+1} [U(P_k, f) - L(P_k, f)] \\ &\quad + \sum_{r=1}^m (M^{(r)} - m^{(r)}) \delta_r \\ &< (m+1) \cdot \frac{\epsilon}{2(m+1)} + 2k \cdot \frac{\epsilon}{4k} \\ &= \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

Case 2: Discontinuities at Endpoints

- If $a = x_1$, use interval $[a, a + \delta_1]$
- If $x_m = b$, use interval $[b - \delta_m, b]$
- Similar arguments apply
- The total contribution from discontinuity intervals remains $< \frac{\epsilon}{2}$

Therefore, in all cases, for any $\epsilon > 0$, there exists a partition P such that:

$$U(P, f) - L(P, f) < \epsilon$$

Corollary: Piecewise Continuous Functions

Corollary. If $f : [a, b] \rightarrow \mathbb{R}$ is piecewise continuous on $[a, b]$, then f is integrable on $[a, b]$.

- Piecewise continuous functions have at most finitely many discontinuities
- They are bounded on each continuous piece
- Therefore, they satisfy the conditions of the theorem

Summary

Note: This result is optimal——functions with infinitely many discontinuities may or may not be integrable.

Example 1: Integrable with Infinitely Many Discontinuities

Thomae's Function (The "Ruler Function")

Definition

Thomae's function is defined on $[0, 1]$ by:

$$f(x) = \begin{cases} \frac{1}{q} & \text{if } x = \frac{p}{q} \text{ is rational in lowest terms,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

- **Discontinuities:** Discontinuous at **every rational number** in $[0, 1]$.
- **Set of discontinuities:** $\mathbb{Q} \cap [0, 1]$, which is **countable**
- **Bounded:** $0 \leq f(x) \leq 1$.
- **Integrability:** f is **Riemann integrable** on $[0, 1]$.
- Moreover, $\int_0^1 f = 0$.

Example 2: NOT Integrable with Infinitely Many Discontinuities

Dirichlet Function

Definition

The Dirichlet function is defined on $[a, b]$ by:

$$g(x) = \begin{cases} 1 & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

- **Discontinuities:** Discontinuous at **every point** in $[a, b]$.
- **Set of discontinuities:** The entire interval $[a, b]$,
- **Bounded:** $0 \leq g(x) \leq 1$.
- **Integrability:** **not Riemann integrable.**

Example 1: Signum Function

Example 1. Let $f(x) = \operatorname{sgn} x$, $x \in [-2, 2]$, where:

$$f(x) = \begin{cases} -1 & -2 \leq x < 0 \\ 0 & x = 0 \\ 1 & 0 < x \leq 2 \end{cases}$$

- f is bounded on $[-2, 2]$ since $|f(x)| \leq 1$
- f is continuous except at only one point ($x = 0$)
- Therefore, by Theorem 11.5.3, f is integrable on $[-2, 2]$

Example 2: Floor Function

Example 2. Let $f(x) = [x]$ (floor function), $x \in [0, 2]$, where:

$$f(x) = \begin{cases} 0 & 0 \leq x < 1 \\ 1 & 1 \leq x < 2 \\ 2 & x = 2 \end{cases}$$

- f is bounded on $[0, 2]$ since $|f(x)| \leq 2$
- f is continuous except at points $x = 1, 2$
- Finite discontinuities $\rightarrow$ integrable on $[0, 2]$

Example 3: Monotone Increasing Function

Example 3. Let f be defined on $[0, 1]$ by:

$$f(0) = 0 \quad \text{and} \quad f(x) = \frac{1}{2^{n-1}}, \quad \frac{1}{2^n} < x \leq \frac{1}{2^{n-1}}$$

for $n = 1, 2, 3, \dots$

- f is monotone increasing on $[0, 1]$
- f is bounded ($0 \leq f(x) \leq 1$)
- By Theorem 11.5.1, all monotone functions are integrable
- Therefore, f is integrable on $[0, 1]$

Example 4: Function with Countably Many Discontinuities

Example 4. Let f be defined on $[0, 1]$ by:

$$f(0) = 0 \quad \text{and} \quad f(x) = (-1)^{r-1}, \quad \frac{1}{r+1} < x \leq \frac{1}{r}$$

for $r = 1, 2, 3, \dots$

- f is bounded on $[0, 1]$ since $|f(x)| \leq 1$
- Discontinuous at points $0, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$
- These discontinuities have only one limit point (0)
- Still integrable because bounded with countable discontinuities

Key Insights

- Boundedness + finite discontinuities $\rightarrow$ integrability
- Monotonicity $\rightarrow$ integrability (even with countably many discontinuities)
- Some functions with countably many discontinuities are still integrable
- The set of discontinuities must be "small enough" (measure zero)

These examples demonstrate the power of the Riemann integrability theorems.

Example 1: Integrable Case

$$f(x) = \begin{cases} 1, & \text{if } x = \frac{1}{n}, n \in \mathbb{N}, \\ 0, & \text{otherwise,} \end{cases} \quad x \in [0, 1].$$

- f is discontinuous at countably many points $\{1, \frac{1}{2}, \frac{1}{3}, \dots\}$.
- Continuous everywhere else.
- Discontinuities form a set of measure zero.

$$\therefore f \in \mathcal{R}[0, 1], \quad \int_0^1 f(x) dx = 0.$$

Example 2: Non-Integrable Case

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \cap [0, 1], \\ 0, & \text{if } x \in \mathbb{I} \cap [0, 1]. \end{cases}$$

- This is the **Dirichlet function**.
- Discontinuous at every point in $[0, 1]$.

$$\therefore f \notin \mathcal{R}[0, 1].$$

Conclusion

- Infinitely many discontinuities $\nRightarrow$ not integrable.
- What matters is the **size of the set of discontinuities**.
- Measure zero $\Rightarrow$ integrable.
- Positive measure $\Rightarrow$ not integrable.